

Customer Purchase Prediction and Personalized Product Recommendation System

1. Problem Statement

E-commerce platforms offer thousands of products, making it difficult for customers to find products that match their interests and needs. Traditional recommendation systems often rely mainly on past user-item interactions and may not sufficiently consider a customer's recent behavior or purchase intention.

Therefore, there is a need for an intelligent system that can analyze customer behavior, predict the likelihood of a purchase, identify customers with similar shopping patterns, and provide personalized product recommendations.

2. Case-Based Scenario

Consider a customer visiting an e-commerce website to purchase a laptop.

During the session, the customer:

- Views several laptops.
- Repeatedly views one particular laptop.
- Adds the laptop to the cart.
- Looks at related laptop accessories.
- Does not immediately complete the purchase.

The system now needs to determine:

Which products should be recommended to this customer?

Simply showing popular products may not be useful because the customer is clearly interested in laptops and related products.

Instead, the system should analyze the customer's interactions and determine their purchase intention. Based on this analysis, it can recommend products that are more relevant to the customer's current interests and behavior.

3. Proposed Solution

Our project, **Customer Purchase Prediction and Personalized Product Recommendation System**, uses Machine Learning to analyze customer-product interactions and generate personalized recommendations.

The system uses information such as:

- Product views
- Add-to-cart actions
- Previous purchases
- Interaction frequency
- Product engagement
- Recent customer activity

The raw interaction data is processed and converted into meaningful behavioral features. These features are then used by Machine Learning models to predict purchase likelihood and support personalized recommendations.

4. System Workflow

Step 1 – Customer Behavior Analysis

The system analyzes the customer's interaction history, including products viewed, products added to the cart, and previous purchases.

This helps the system understand the customer's interests and level of engagement.

Step 2 – Data Preprocessing and Feature Engineering

The raw e-commerce event data is cleaned and transformed into features that can be used by Machine Learning models.

Examples include:

- Number of product views
- Number of cart additions
- Previous purchase activity
- Interaction frequency
- Product and category information

- Recent activity

Step 3 – Purchase Prediction

The system treats purchase prediction as a **binary classification problem**.

For a product viewed by a customer, the model predicts whether the customer is likely to purchase that product within a defined time period.

We experimented with Machine Learning models such as:

- Logistic Regression
- HistGradientBoosting

Logistic Regression was used as a baseline model, while HistGradientBoosting was selected as the final purchase prediction model because it provided better validation performance.

Step 4 – Customer Segmentation

Customers can be grouped according to similarities in their shopping behavior.

For example, customers with similar levels of product views, cart activity, and purchase behavior can be placed into the same behavioral group.

This helps the recommendation system understand different types of customers.

Step 5 – Purchase Journey Identification

The system can interpret the customer's current stage in the purchase journey:

Exploring → Considering → Ready to Purchase

For example:

- **Exploring:** The customer is browsing and viewing multiple products.
- **Considering:** The customer is repeatedly interacting with specific products or adding products to the cart.
- **Ready to Purchase:** The customer's behavior indicates stronger purchase intent.

The type of recommendation can be adjusted according to the customer's current stage.

Step 6 – Personalized Recommendation

The recommendation component uses multiple signals, including:

Customer Preferences + Behavioral Activity + Purchase Probability + Customer Segment + Purchase Journey

These signals are used to identify products that are relevant to the customer.

For a new customer with little or no interaction history, an initial interest-selection mechanism can provide preference information and help address the **cold-start problem**.

Step 7 – Explainable Recommendation

The system can provide a simple explanation along with a recommendation.

For example:

"Recommended because you frequently viewed products from this category."

This makes the recommendation more transparent and easier for the customer to understand.

5. Case Outcome

Suppose the customer repeatedly views a particular laptop and adds it to their cart.

The system recognizes these interactions as stronger indicators of purchase intent. The purchase prediction model then generates a purchase probability for the product.

Based on this information, the recommendation system can prioritize relevant products, such as:

- Similar laptops
- Related accessories
- Products from the same category
- Other products matching the customer's interests

The recommendations can also change according to the customer's purchase journey.

For example:

- During the **Exploring** stage, the system can provide a broader range of relevant products.
 - During the **Considering** stage, it can provide more closely related products.
 - During the **Ready to Purchase** stage, it can prioritize products with stronger predicted purchase likelihood.
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6. Overall System Pipeline

The proposed system follows the following workflow:

Customer Interaction Data

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Data Preprocessing

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Behavioral Feature Engineering

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Purchase Prediction

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Customer Segmentation

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Purchase Journey Identification

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Personalized Recommendation

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Explainable Recommendation

7. Expected Results

The proposed system aims to:

1. Predict whether a customer is likely to purchase a product.
2. Analyze customer shopping behavior.
3. Identify customers with similar behavioral patterns.
4. Generate personalized product recommendations.
5. Adapt recommendations according to the customer's purchase journey.
6. Compare Machine Learning models using appropriate evaluation metrics.

7. Provide understandable explanations for recommendations.
 8. Present predictions and recommendations through an interactive interface.
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8. Conclusion

The **Customer Purchase Prediction and Personalized Product Recommendation System** combines customer behavior analysis, Machine Learning, and personalized recommendation techniques to improve product discovery on e-commerce platforms.

Instead of relying only on general popularity or historical interactions, the proposed system considers the customer's current behavior and predicted purchase intent. This allows the system to provide recommendations that are more relevant to the customer's current shopping activity and purchase journey.